

	$ \frac{3}{2}, \frac{3}{2}\rangle$	$ \frac{3}{2}, \frac{1}{2}\rangle$	$ \frac{3}{2}, -\frac{1}{2}\rangle$	$ \frac{3}{2}, -\frac{3}{2}\rangle$	$ \frac{1}{2}, \frac{1}{2}\rangle_1$	$ \frac{1}{2}, -\frac{1}{2}\rangle_1$	$ \frac{1}{2}, \frac{1}{2}\rangle_2$	$ \frac{1}{2}, -\frac{1}{2}\rangle_2$
$\langle \frac{3}{2}, \frac{3}{2} $	$\frac{3J}{4}$							
$\langle \frac{3}{2}, \frac{1}{2} $	0	$\frac{3J}{4}$						
$\langle \frac{3}{2}, -\frac{1}{2} $	0	0	$\frac{3J}{4}$					
$\langle \frac{3}{2}, -\frac{3}{2} $	0	0	0	$\frac{3J}{4}$				
$\langle \frac{1}{2}, \frac{1}{2} $	$\frac{i(2d_{12}-d_{13}-d_{23})(e_x+ie_y)}{4\sqrt{2}}$	$\frac{i(2d_{12}-d_{13}-d_{23})e_z}{2\sqrt{6}}$	$\frac{i(2d_{12}-d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{6}}$	0	$-\frac{3J}{4}$			
$\langle \frac{1}{2}, -\frac{1}{2} $	0	$\frac{i(2d_{12}-d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{6}}$	$\frac{i(2d_{12}-d_{13}-d_{23})e_z}{2\sqrt{6}}$	$\frac{(2d_{12}-d_{13}-d_{23})}{4\sqrt{2}}$	0	$-\frac{3J}{4}$		
$\langle \frac{1}{2}, \frac{1}{2} $	$\frac{\sqrt{3}(d_{13}-d_{23})(e_x+ie_y)}{4\sqrt{2}}$	$\frac{i(d_{13}-d_{23})e_z}{2\sqrt{2}}$	$-\frac{i(d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{2}}$	0	$\frac{i(d_{12}+d_{13}+d_{23})}{2\sqrt{3}}$	$-\frac{i(d_{12}+d_{13}+d_{23})(e_x-ie_y)}{2\sqrt{3}}$	$-\frac{3J}{4}$	
$\langle \frac{3}{2}, \frac{3}{2} $	0	$\frac{i(d_{13}-d_{23})(e_x+ie_y)}{4\sqrt{2}}$	$\frac{i(d_{13}-d_{23})e_z}{2\sqrt{2}}$	$-\frac{i\sqrt{3}(d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{2}}$	$-\frac{i(d_{12}+d_{13}+d_{23})(e_x+ie_y)}{2\sqrt{3}}$	$-\frac{ie_z(d_{12}+d_{13}+d_{23})}{2\sqrt{3}}$	0	$-\frac{3J}{4}$